IAN SNIDER

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Software engineer skilled in autonomous control systems, HPC, machine learning, and radiation safety, seeking positions in control system design and embedded systems for robotics with an emphasis on safety critical architecture.

EXPERIENCE

Lawrence Berkeley National Laboratory - *Lab Affiliate*, *Berkeley, CA* — *Aug 2025 - present*

- Graduate researcher in the Bay Area Neutron Group studying experimental techniques for indirect cross-section measurement of short-lived nuclei

Lawrence Livermore National Laboratory - *Graduate Intern*, *Livermore, CA* — *June 2025 - present*

- Developed CRISP (CRItical Simulation Pipeline) for parallelizing jobs from the radiation transport code: COG
- Test driven Python development included Poetry, CI/CD, SQLite, MPI, CLI, feature branching, and Sphinx
- Used 2 world-class LLNL HPC clusters Dane and Ruby with Slurm sbatch/srun scripts
- CRISP sucessfully automated the materials compiling and k_{eff} benchmarking of ~3420 ICSBEP critical reactors

Brookhaven National Laboratory - *Nuclear Science Intern*, *Upton, NY* — *Summers 2022 - 2024*

- Developed BRR (Bayesian Resonance Reclassifier) for resonance reclassification on National Nuclear Data Center clusters
- Applied machine learning using Scikit-learn and Pytorch to classify resonance spin-group assignments for capture cross-sections of In-115, Pb-206, and U-238
- Used Random Matrix Theory and nuclear data to create synthetic training sets
- Used the neutron transport code OpenMC to perform perturbative sensitivity analyses of the thermal $1/v$ neutron capture cross-sections for critical reactors on Gd-155, 157 and U-235, 238

Truman State University - *Student Astronomy Researcher*, *Kirksville, MO* — *2021 - 2022*

- Calculated trajectories for ~3000 Starlink satellites to optimize telescope viewing plans and developed a GUI

PROJECTS

WashU Robotics MATE ROV & MARINER Projects - ***Mechanical Lead*** — *2023 - 2025*

- Collaborated with other students to develop an advanced autonomous underwater vehicle (AUV)
- Prototyped hydrodynamic dive control using an IMU and sonar state estimation with a Kalman Filter
- Built a chassis and buoyancy engine

ESE 4481 Autonomous Aerial Vehicle Control - ***Student*** — *Spring 2025*

- Designed and succesfully implemented a cascaded PID/PD control system on a CrazyFlie 2.1 quadcopter

Society of Physics Students - ***Demo Chair*** — *2020 - 2023*

- Developed 3 new physics demos, performed demos at meetings, and provided volunteer physics tutoring
- Wrote and proctored exams for 2022 & 2023 Science Olympiads (“Crave the Wave” and “Remote Sensing”)

SKILLS

- **Programming:** C, C++, Python, Go, Bash, Lua, MATLAB, Mathematica, Octave, LaTeX, Typst, Vimscript
- **Software/Technical:** Simulink, SolidWorks, OpenMC, COG, Pytorch, Scikit-learn, Slurm, Git, Linux, Arduino
- **Physics/engineering:** Autonomous Aerial Vehicle Control, Classical Mechanics, Electrodynamics, Electronics, Vibrations, Quantum Mechanics, Mathematical Physics, Nuclear Physics, Fluid Mechanics, Solid Mechanics, Acoustics, Materials Science, Thermal Systems, Turbojets, Ramjets
- **Math:** Linear Algebra, ODEs, Computing Structures, Control Systems, Machine Learning, and Optimizations

EDUCATION

Graduate Researcher in Nuclear Engineering - University of California, Berkeley		*Aug 2025 - present*
B.S. Mechanical Engineering - Washington University in St. Louis	4.00/4.00	*May 2025*
B.A. Physics - Truman State University	3.89/4.00	*Dec 2024*